

Vancomycin exposure — a mini report

Your Name

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1 Introduction

A single intravenous dose of a drug that follows first-order elimination is described by Equation 1, where D is the dose, V_d the volume of distribution and k_e the elimination rate constant.

$$C(t) = \frac{D}{V_d} e^{-k_e t} \quad (1)$$

2 Methods

Concentrations were read from `data/patient-concentrations.csv`. The elimination rate constant was estimated by least squares on the log-transformed concentrations, and the half-life followed from Equation 2.

$$t_{1/2} = \frac{\ln 2}{k_e} \quad (2)$$

3 Results

Figure 1 shows the observed concentrations with the fitted curve. The fitted initial concentration is 19.9 mg/L and the estimated half-life is 3.1 h.

4 Summary

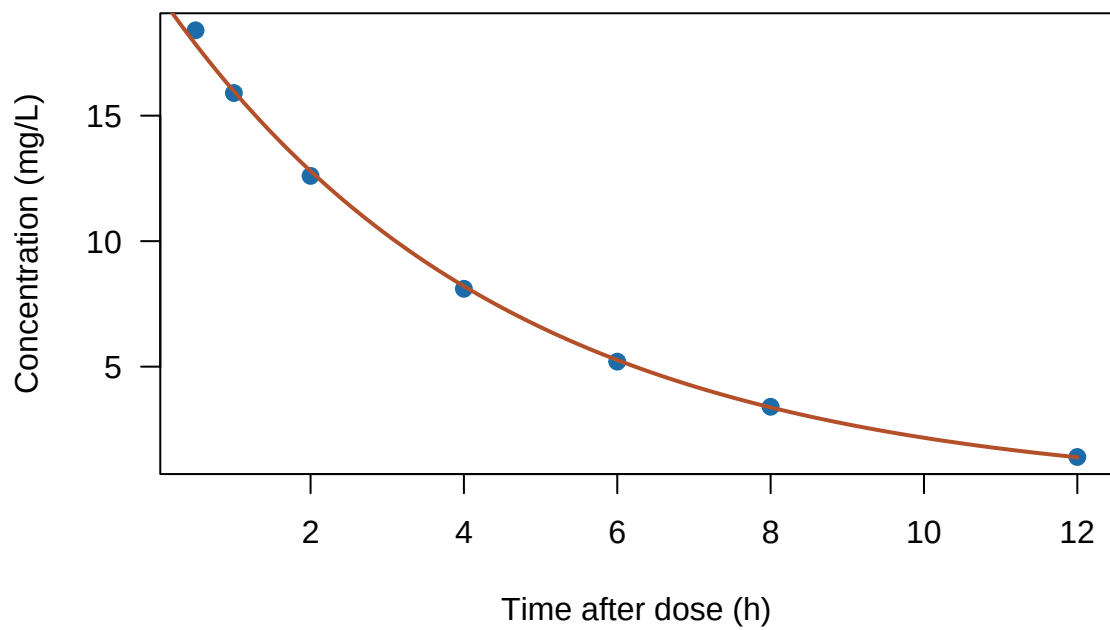


Figure 1: Observed concentrations and the fitted mono-exponential decay.

Table 1: Fitted parameters

Quantity	Estimate
C_0 (mg/L)	19.9
k_e (h^{-1})	0.222
$t_{1/2}$ (h)	3.1

The fitted values in Table 1 are computed at render time, so they cannot drift out of step with the figure in Figure 1.